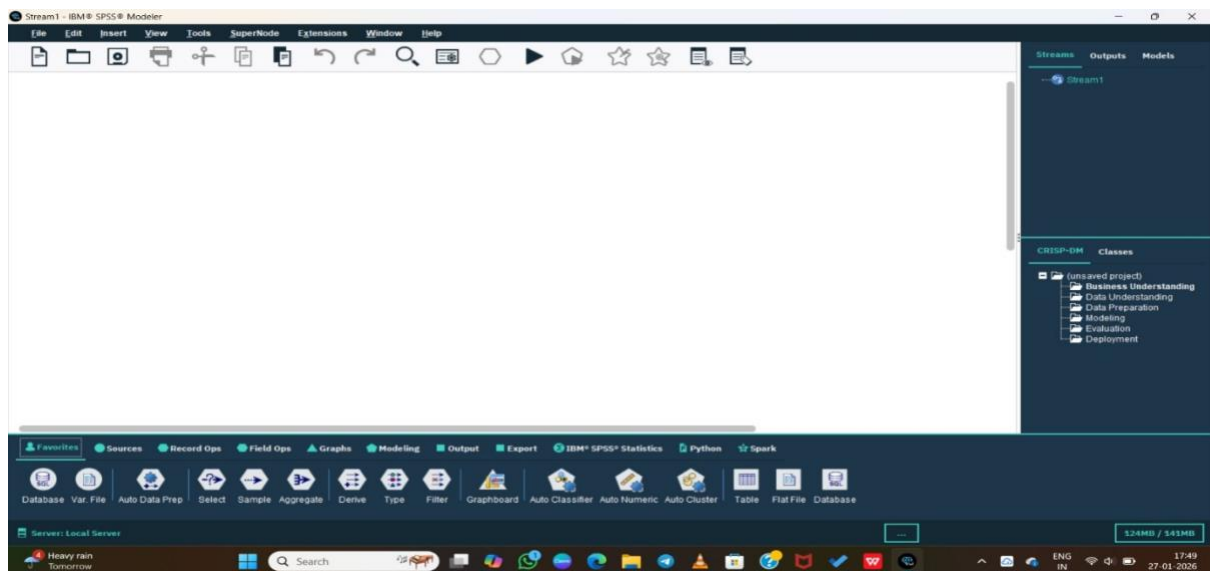


# Create a data mining project to predict churn in telecommunication.

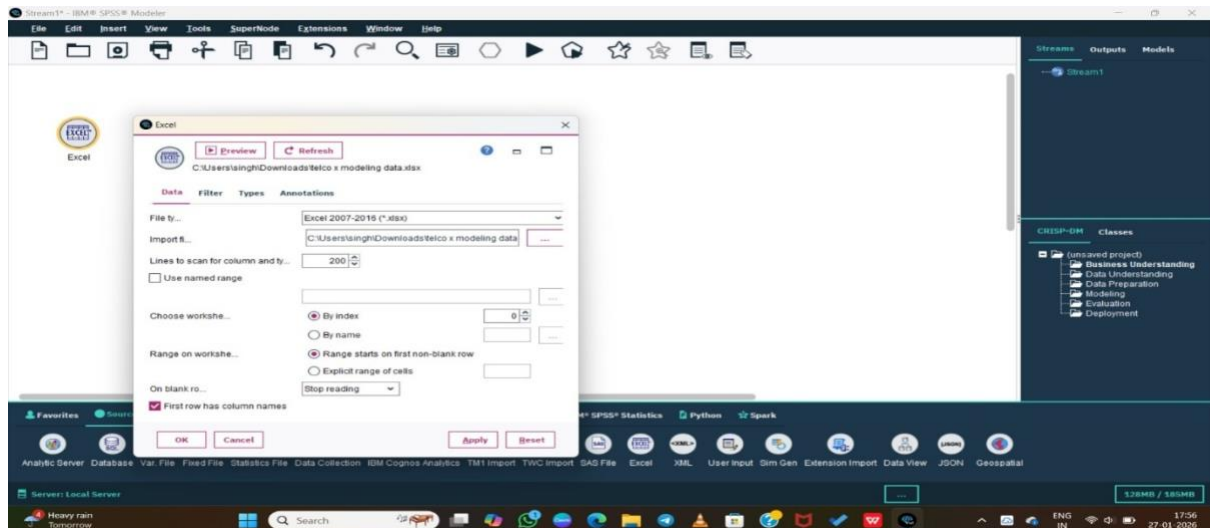
(Telco x modelling data.xlsx)

**Step:1** Start the IBM SPSS modeler and set the working folder.



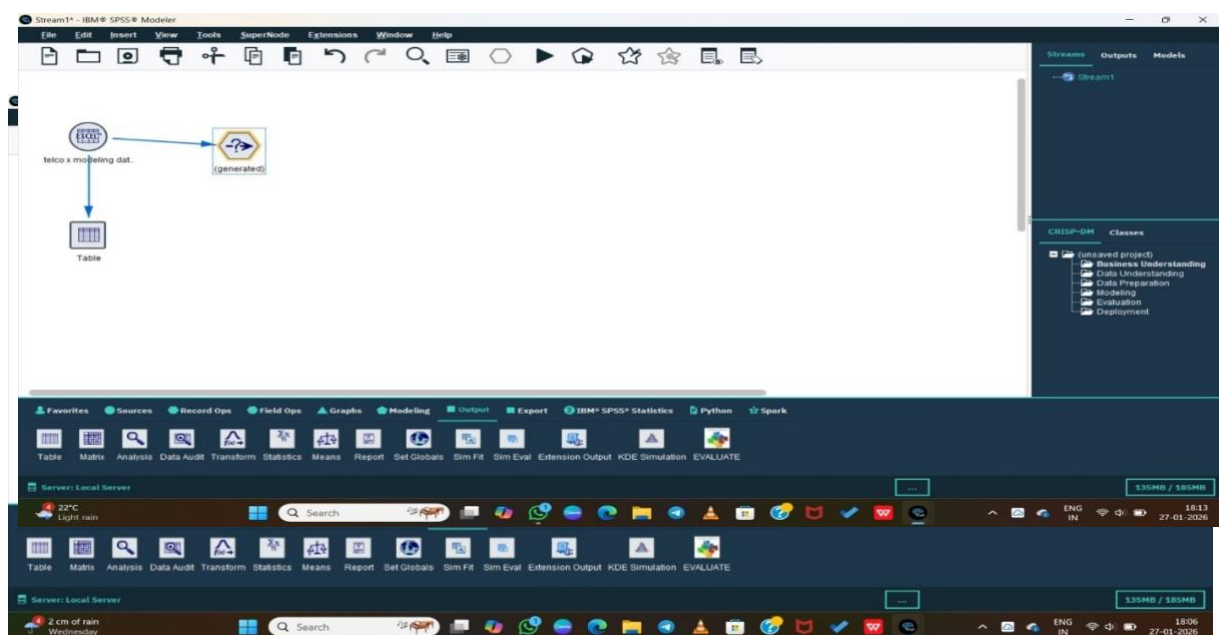
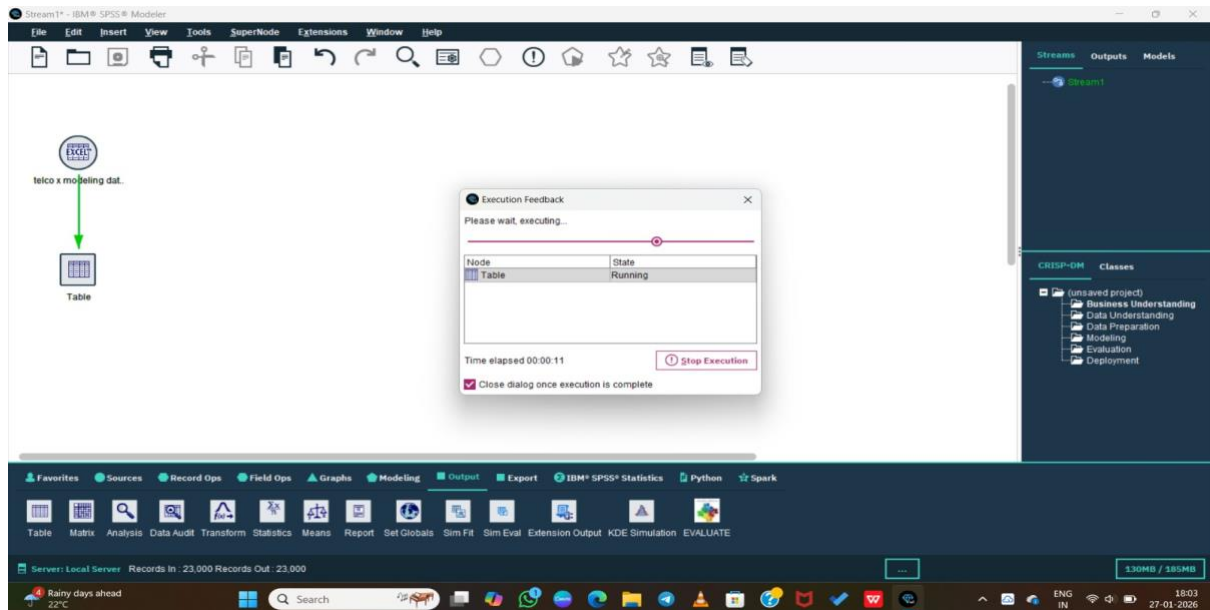
**Step:2** Then go to sources palette, place an excel node and edit the excel node and set the

- import file field to open (telco x modelling data.xlsx)
- The preview button let's you examine the first 10 seconds of the data set.



**Step:3** Then go to the output palette and select table node and wait for loading.

- You will see all the records in the table then select on
- Then in the table go to option generate and select node (“And”) and click ok to close the table output window.
- A select node named (generated ) is placed in the upper left corner of the stream canvas.
- Insert the generated node between the excel node and table node.

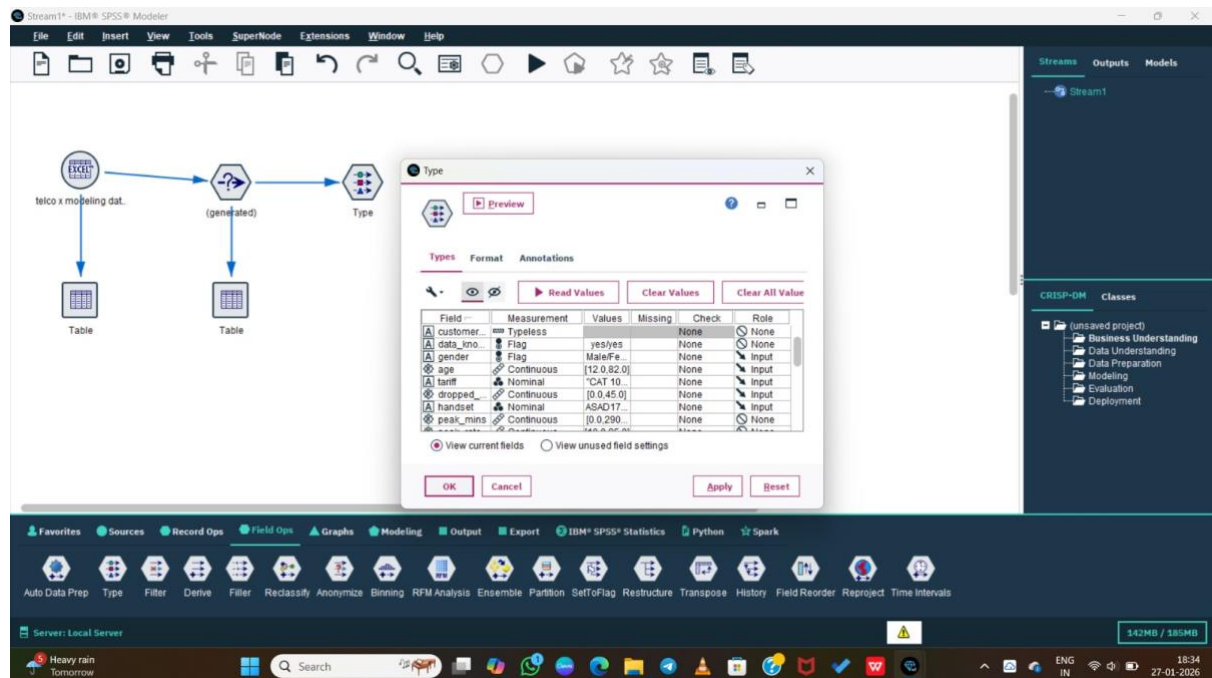


**The result appear as follows.**

**Step:4** Build a model using historical data.

- The next step is that you will use a model to predict churn.it is essential for modelling that predictors are selected and that the target field is specified.

- From the field ops palette, add a type node downstream from the data known select node and edit the type node.



**Step:5** From the modeling palette, add a chaid node downstream from the type node.

- The CHAID node is labelled with the target field, churn.
- Run the CHAID node.

Stream1\* - IBM® SPSS® Modeler

File Edit Insert View Tools SuperNode Extensions Window Help

telco x modeling dat. (generated) Type chum

Table Table

Execution Feedback

Please wait, executing...

| Node | State        |
|------|--------------|
| chum | 16% complete |

Time elapsed 00:00:05

Stop Execution

Close dialog once execution is complete

Streams Outputs Models

Stream1

CRISP-DM Classes

- (unsaved project)
- Business Understanding
- Data Understanding
- Data Preparation
- Modeling
- Evaluation
- Deployment

Favorites Sources Record Ops Field Ops Graphs Modeling Output Export IBM® SPSS® Statistics Python Spark

AI Supervised Association Segmentation

Cox SVM LSM Bayes Net SLRM Apriori Carma Sequence K-Means Kohonen TwoStep-AS TwoStep Anomaly KNN Isotonic-AS Extension Model XGBoost Linear XGBoost Tree CH4ID Quest Gaussian Mixture KDE Mod

Server: Local Server Elapsed Time 0 hour(s): 0 minute(s): 5 second(s)

144MB / 185MB

Trending videos All eyes on 'Ave...

Search

23°C Light rain

18:39 27-01-2026

Stream1\* - IBM® SPSS® Modeler

File Edit Insert View Tools SuperNode Extensions Window Help

telco x modeling dat. (generated) Type chum

Table Table

chum

Streams Outputs Models

chum

CRISP-DM Classes

- (unsaved project)
- Business Understanding
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Favorites Sources Record Ops Field Ops Graphs Modeling Output Export IBM® SPSS® Statistics Python Spark

AI Supervised Association Segmentation

Cox SVM LSM Bayes Net SLRM Apriori Carma Sequence K-Means Kohonen TwoStep-AS TwoStep Anomaly KNN Isotonic-AS Extension Model XGBoost Linear XGBoost Tree CH4ID Quest Gaussian Mixture KDE Mod

Server: Local Server

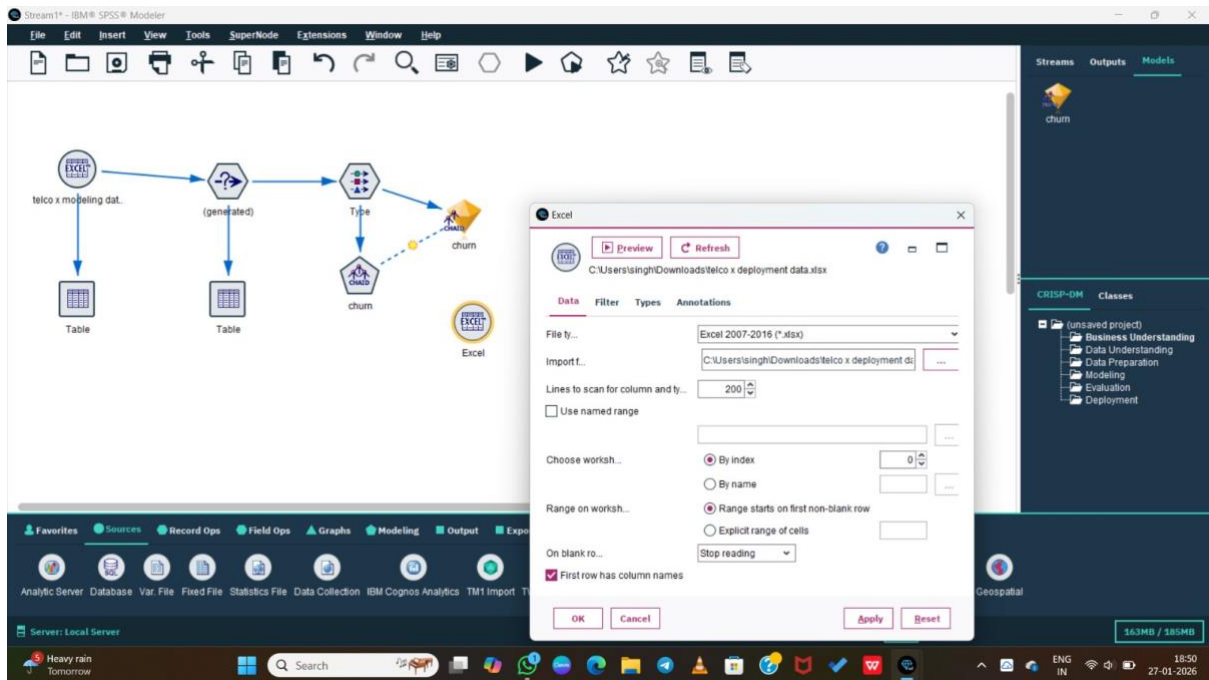
163MB / 185MB

23°C Light rain

18:43 27-01-2026

# [Telco x deployment data.xlsx]

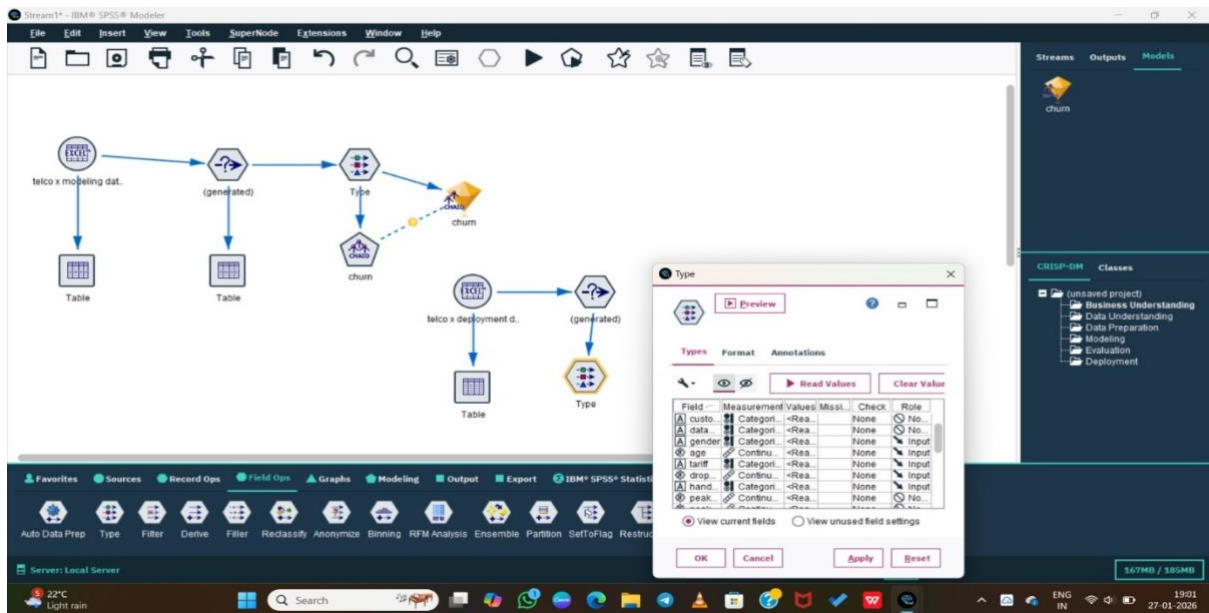
**Step:1** Go to sources palette and add excel node and import data in it as same as did in 1<sup>st</sup> data.



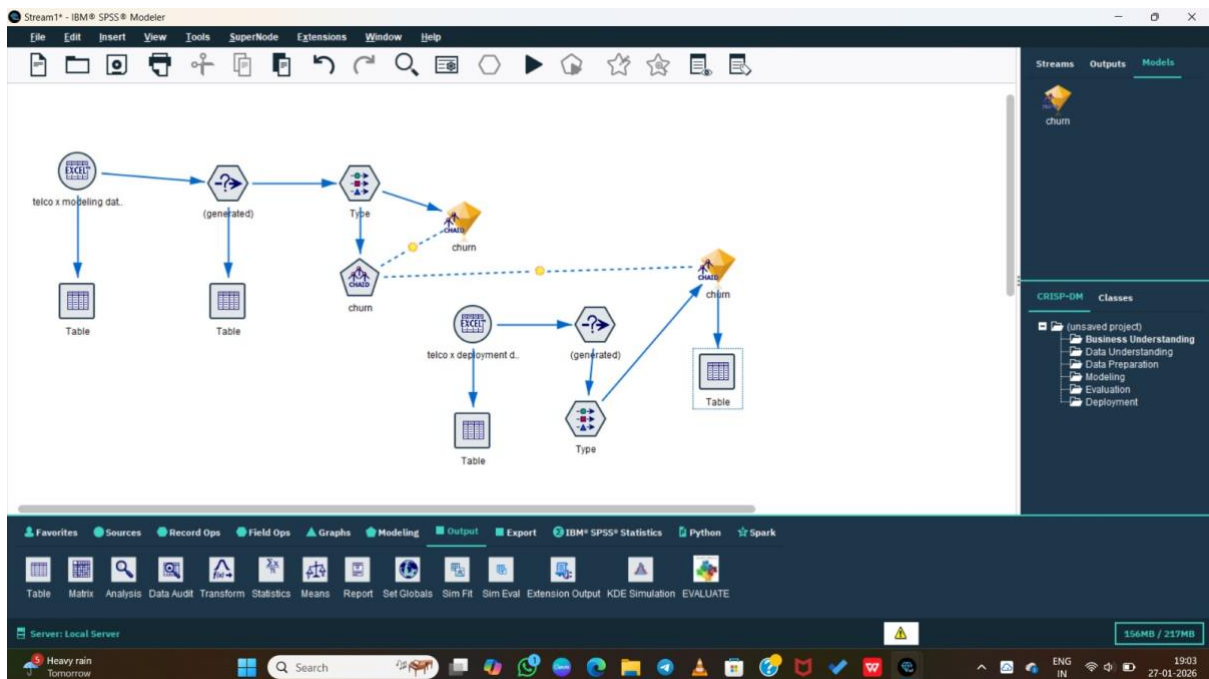
- Then add table from the output palette and select any yes from it.and select node (And) from generate button and then click ok.the the generated node will appear you on the stream canvas.

**Step:2** Then add type palette from field ops option and read values.





- Then copy node the churn and then select table palette and run. The churned data will be appeared.



| Table (23 fields, 9,994 records) #1 |       |           |              |              |              |          |          |          |          |                   |                   |                   |                   |           |
|-------------------------------------|-------|-----------|--------------|--------------|--------------|----------|----------|----------|----------|-------------------|-------------------|-------------------|-------------------|-----------|
| File Edit Generate                  |       |           |              |              |              |          |          |          |          |                   |                   |                   |                   |           |
| Table Annotations                   |       |           |              |              |              |          |          |          |          |                   |                   |                   |                   |           |
|                                     | rate  | bill_peak | offpeak_mins | offpeak_rate | bill_offpeak | gadget_A | gadget_B | gadget_C | gadget_D | gadget_A_revenues | gadget_B_revenues | gadget_C_revenues | gadget_D_revenues | SRC-churn |
| 1                                   | 0.000 | 59.040    | 19.200       | 5.000        | 0.940        | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Active    |
| 2                                   | 0.000 | 20.430    | 36.900       | 5.000        | 1.845        | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Churned   |
| 3                                   | 0.000 | 99.640    | 122.400      | 5.000        | 6.120        | T        | T        | T        | T        | 11.000            | 20.000            | 25.000            | 0.000             | Active    |
| 4                                   | 0.000 | 133.560   | 396.900      | 5.000        | 19.845       | T        | F        | T        | T        | 10.000            | 24.000            | 0.000             | 41.000            | Churned   |
| 5                                   | 0.000 | 84.480    | 300.600      | 5.000        | 15.030       | T        | T        | T        | T        | 8.000             | 23.000            | 33.000            | 40.000            | Active    |
| 6                                   | 0.000 | 84.150    | 432.900      | 2.000        | 5.655        | T        | T        | T        | T        | 0.000             | 23.000            | 26.000            | 43.000            | Active    |
| 7                                   | 0.000 | 124.350   | 729.900      | 5.000        | 36.495       | T        | F        | T        | T        | 9.000             | 19.000            | 0.000             | 37.000            | Churned   |
| 8                                   | 0.000 | 234.600   | 101.400      | 5.000        | 5.070        | T        | F        | T        | T        | 14.000            | 17.000            | 0.000             | 39.000            | Churned   |
| 9                                   | 0.000 | 97.680    | 48.300       | 5.000        | 2.415        | T        | T        | T        | T        | 10.000            | 17.000            | 26.000            | 40.000            | Active    |
| 10                                  | 0.000 | 72.180    | 45.300       | 5.000        | 2.245        | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Churned   |
| 11                                  | 0.000 | 92.740    | 195.300      | 5.000        | 9.915        | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Churned   |
| 12                                  | 0.000 | 16.050    | 464.400      | 5.000        | 23.220       | F        | T        | F        | F        | 0.000             | 0.000             | 26.000            | 0.000             | Active    |
| 13                                  | 0.000 | 121.650   | 470.700      | 2.000        | 9.414        | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Churned   |
| 14                                  | 0.000 | 30.600    | 540.600      | 5.000        | 27.030       | T        | T        | T        | T        | 14.000            | 0.000             | 30.000            | 0.000             | Churned   |
| 15                                  | 0.000 | 80.550    | 129.000      | 5.000        | 4.450        | T        | T        | T        | T        | 11.000            | 22.000            | 29.000            | 37.000            | Active    |
| 16                                  | 0.000 | 51.150    | 548.100      | 2.000        | 10.962       | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Active    |
| 17                                  | 0.000 | 76.200    | 341.800      | 2.000        | 7.236        | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Active    |
| 18                                  | 0.000 | 71.880    | 338.400      | 5.000        | 16.920       | T        | T        | T        | F        | 6.000             | 20.000            | 28.000            | 0.000             | Active    |
| 19                                  | 0.000 | 63.720    | 348.900      | 5.000        | 17.445       | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Active    |
| 20                                  | 0.000 | 9.150     | 403.900      | 5.000        | 20.175       | T        | F        | F        | F        | 9.000             | 23.000            | 0.000             | 0.000             | Active    |
| 21                                  | 0.000 | 13.500    | 194.100      | 5.000        | 9.705        | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Churned   |
| 22                                  | 0.000 | 148.200   | 502.200      | 5.000        | 25.110       | T        | T        | T        | F        | 9.000             | 20.000            | 32.000            | 0.000             | Active    |
| 23                                  | 0.000 | 85.530    | 295.500      | 5.000        | 14.925       | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Active    |
| 24                                  | 0.000 | 60.750    | 331.200      | 5.000        | 16.940       | T        | T        | T        | T        | 0.000             | 21.000            | 31.000            | 37.000            | Churned   |
| 25                                  | 0.000 | 57.600    | 179.500      | 5.000        | 8.775        | F        | F        | T        | T        | 0.000             | 0.000             | 0.000             | 36.000            | Active    |
| 26                                  | 0.000 | 27.810    | 26.700       | 5.000        | 1.335        | F        | T        | T        | T        | 0.000             | 0.000             | 26.000            | 40.000            | Active    |
| 27                                  | 0.000 | 79.380    | 245.200      | 5.000        | 13.240       | T        | F        | F        | F        | 10.000            | 0.000             | 0.000             | 0.000             | Active    |
| 28                                  | 0.000 | 84.150    | 301.200      | 2.000        | 6.024        | F        | F        | F        | F        | 10.000            | 0.000             | 0.000             | 0.000             | Active    |
| 29                                  | 0.000 | 58.650    | 231.000      | 5.000        | 11.550       | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Churned   |
| 30                                  | 0.000 | 29.850    | 627.000      | 2.000        | 12.940       | T        | F        | F        | F        | 14.000            | 19.000            | 0.000             | 0.000             | Active    |
| 31                                  | 0.000 | 53.440    | 342.000      | 5.000        | 17.100       | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Churned   |
| 32                                  | 0.000 | 143.340   | 29.100       | 5.000        | 1.455        | T        | T        | F        | F        | 0.000             | 20.000            | 27.000            | 0.000             | Churned   |
| 33                                  | 0.000 | 85.240    | 224.100      | 5.000        | 11.205       | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Active    |
| 34                                  | 0.000 | 80.250    | 450.000      | 2.000        | 9.000        | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Active    |
| 35                                  | 0.000 | 77.740    | 475.800      | 5.000        | 23.790       | T        | T        | T        | F        | 0.000             | 23.000            | 25.000            | 0.000             | Active    |
| 36                                  | 0.000 | 93.420    | 147.100      | 5.000        | 8.355        | T        | T        | T        | T        | 0.000             | 23.000            | 30.000            | 38.000            | Churned   |
| 37                                  | 0.000 | 53.400    | 253.200      | 5.000        | 12.660       | T        | F        | F        | F        | 0.000             | 19.000            | 0.000             | 0.000             | Churned   |
| 38                                  | 0.000 | 79.830    | 102.000      | 5.000        | 5.100        | F        | F        | F        | F        | 0.000             | 0.000             | 0.000             | 0.000             | Active    |
| 39                                  | 0.000 | 99.300    | 452.700      | 5.000        | 22.635       | T        | T        | T        | T        | 0.000             | 20.000            | 31.000            | 39.000            | Churned   |
| 40                                  | 0.000 | 81.840    | 458.700      | 5.000        | 24.450       | F        | F        | F        | F        | 0.000             | 20.000            | 24.000            | 39.000            | Churned   |

3 cm of rain  
Tonight

Search

19:06  
27-01-2026